

KPI Framework, Business Experiment Analysis & Decision Recommendation

Business Context

A subscription-based digital product company conducted an A/B experiment to evaluate a new onboarding and activation campaign. Users were randomly assigned to either the Control group (existing onboarding experience) or the Treatment group (new onboarding experience). The goal was to determine whether the Treatment should be launched to all users.

Business Problem Statement

The company needs to decide whether the new onboarding campaign should replace the existing onboarding process.

The decision impacts:

- Product Team
- Marketing Team
- Business Leadership
- Customers

The campaign should improve:

- Paid Conversion Rate
- User Engagement
- Revenue

Potential risks include:

- Increased refund requests
- More support tickets
- Reduced engagement
- Lower revenue quality

The recommendation is based on experiment analysis, hypothesis testing, and guardrail metrics.

Dataset Description

The dataset contains user-level experiment data, including:

- User ID
- Experiment Group (Control/Treatment)
- Landing Page Visit

- Trial Started
 - Onboarding Completed
 - Paid Conversion
 - Revenue
 - Refund Request
 - Support Ticket
 - Engagement Score
 - Days to Convert
 - Region
 - Device Type
 - Traffic Source
 - Plan Type
-

North Star Metric

Paid Conversion Rate

This metric measures the percentage of users who successfully convert into paying customers and directly contributes to subscription revenue growth.

KPI Tree Summary

North Star Metric

- Paid Conversion Rate

Primary Drivers

- User Acquisition
- User Activation
- Monetization

Supporting Drivers

- Landing Page Visit Rate
- Trial Start Rate
- Onboarding Completion Rate
- Revenue per User
- Revenue per Converted User

Guardrail Metrics

- Refund Rate
 - Support Ticket Rate
 - Engagement Score
 - Average Days to Convert
-

Experiment Analysis Approach

The dataset was cleaned and validated by checking:

- Missing values
- Duplicate User IDs
- Invalid binary values
- Revenue outliers
- Group distribution
- Segment distribution

Control and Treatment groups were compared across key business metrics.

Hypothesis Test Summary

- Null Hypothesis (H_0): No improvement in Paid Conversion Rate.
- Alternative Hypothesis (H_1): Treatment improves Paid Conversion Rate.
- Test Type: One-tailed Two-Proportion Z-Test
- Significance Level: 5%

The hypothesis test provides statistical evidence to support the business decision.

Guardrail Metrics Considered

- Refund Rate
- Support Ticket Rate
- Average Engagement Score
- Average Days to Convert
- Revenue Quality

These metrics ensure the Treatment improves conversions without negatively affecting customer experience.

Final Recommendation

Based on the experiment results and guardrail analysis, the Treatment onboarding experience is recommended for rollout to all users because it demonstrates improved business performance while maintaining acceptable customer experience.

Assumptions

- Users were randomly assigned to experiment groups.
- Data accurately reflects user behavior.

- Sample size is sufficient for hypothesis testing.
-

Limitations

- Experiment duration is limited.
 - Long-term customer behavior is not measured.
 - External business factors may influence user behavior.
-

Repository Structure

part2_kpi_experiment/

- data/
 - analysis/
 - outputs/
 - screenshots/
 - README.md
-

Screenshots Included

- summary_metrics.png
- hypothesis_test_output.png
- kpi_tree_preview.png